

Tutorial

- You are tasked with designing an person **counter system (count will not be more than 255)** using the 8086 microprocessor and **two 8255 Programmable Peripheral Interface (PPI)** chips.
- Assume that each time a person enters the room, a low to high signal appears at the NMI Pin (need not design the circuit that generates signal at NMI pin).
- Two **8255 PPIs** are connected in I/O-mapped mode.
- **Hardware Configuration:**
 - **8255 #1 Port C (Input):** Reads a control switch connected to PC0.
 - PC0 = 0 → Pause counting and display the last count on 8 bit LED.
 - PC0 = 1– Enable counting and display the last count on 8 bit LED.
 - **8255 #2 Port C (Output):** Displays the 8-bit count value (binary) using LEDs.
 - A reset is used to clear the count.
 - Allowed address range for I/O- F0-F7_H

Memory Interfacing

- ROM

- 64K at 00000_H
- 64k at F0000_H

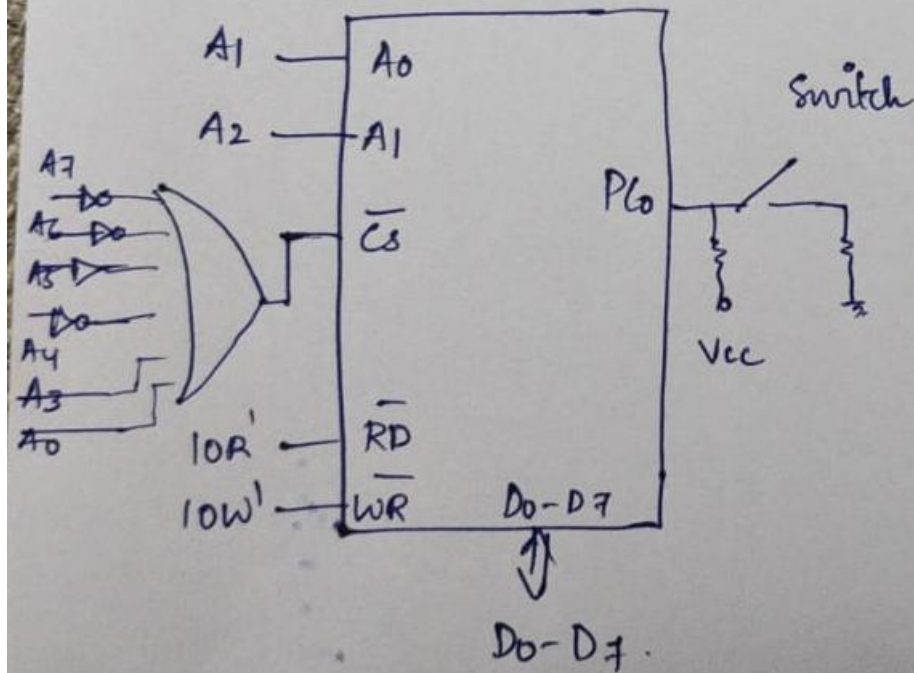
32K chips are available.

- 64 k RAM at C0000_H
- No of decoders available -2
- Assume required no of OR gates, Not gates available

8255

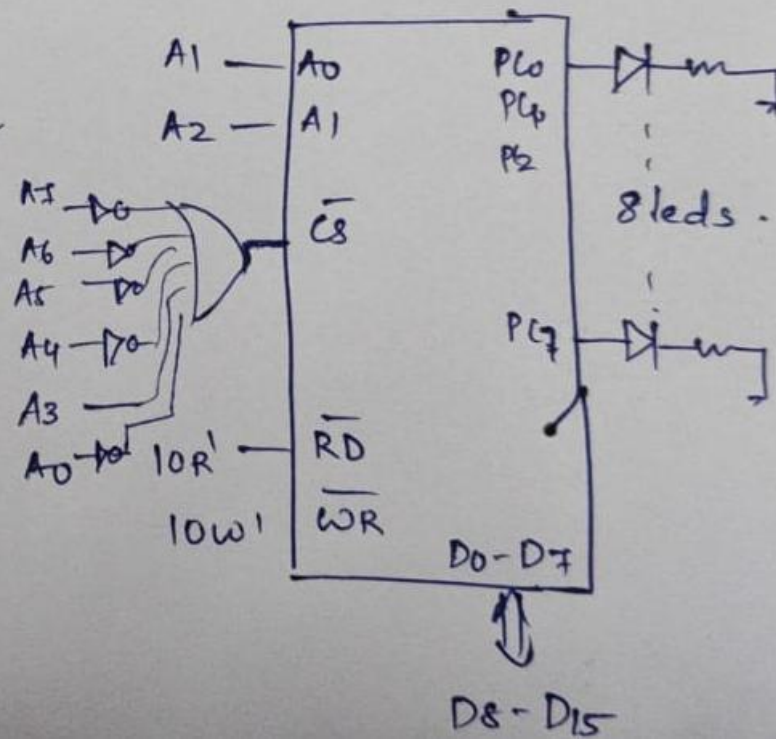
8255 #1

Address	- P0	0000	(PA)
	F2	0010	(PB)
	F4	0100	(PC)
	F6	0110	(PH)



8255 #2

Address	- F1	0001
	- F3	0011
	- F5	0101
	- F7	0111



ISR :

IN AL, F4H

AND AL, 01H

JZ l1

INC CL

l1 : ~~quit~~

MOV AL, CL

OUT F7, AL

IRET

• model tiny

• data

• code

• start up

mov al, ~~80H~~ ~~80H~~ 80H

out F7H, al

mov al, 81H

out ~~F7H~~ F6H, al

mov cl, 0

l2 : jmp l2

• exit

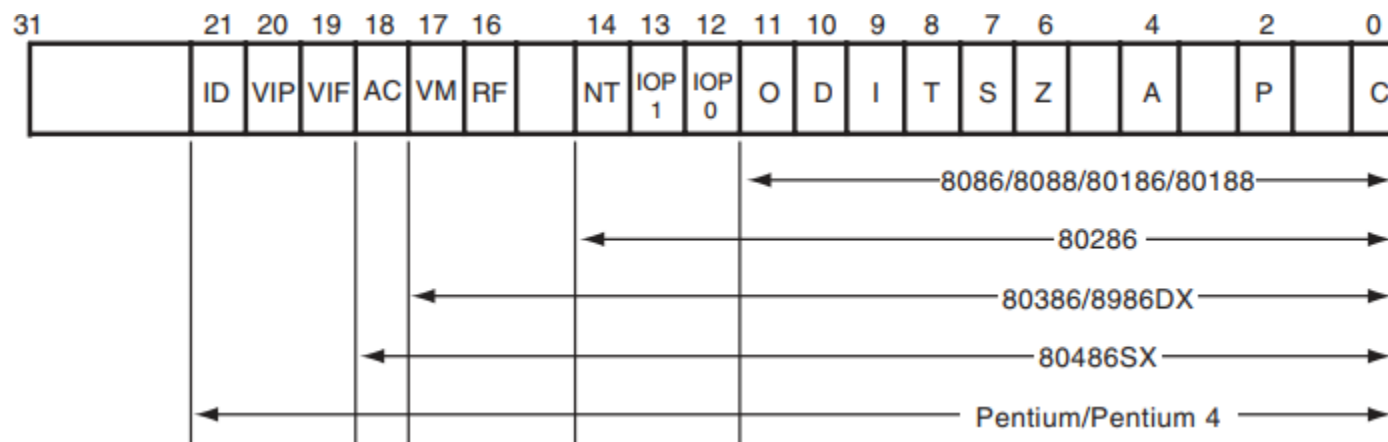
end.

x data
are 0's

CS'	A ₁	A ₀	Selected
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register
1	X	X	8255 Not Selected

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
1 – I/O Mode	Port A Mode 0 0 - Mode 0 0 1 - Mode 1 1 x – Mode2		Port A 1 – i/p 0 – o/p	Port C Upper 1 – i/p 0 – o/p	Port B Mode 0- Mode0 1- Mode1	Port B 1 – i/p 0 – o/p	Port C Lower 1 – i/p 0 – o/p
	Group A				Group B		

- NMI is INT 2
- If flag content is 600_H
- CS and IP at the time of interrupt is 1000 and 2346_H
- Stack is at SS:1208_H
- What is the value of stack after interrupt is raised



1207	06H
1206	00H
1205	10H
1204	00H
1203	23H
1202	46H

- What is the IVT location that will be used to get IP and CS of ISR?